MILITARY SPECIFICATION

REFRIGERATORS, MECHANICAL, BIOLOGICAL, LOW-TEMPERATURE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of mechanically refrigerated, low-temperature, biological refrigerators.

1.2 Classification. Refrigerators covered by this specification shall be of the following types as specified (see 6.2):

Type I	-	Chest type (11-13 cubic foot storage capacity)
Type II	-	Upright type (15-20 cubic foot storage capacity)

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

BB-F-1421	-	Fluorocarbon Refrigerants
FF-C-77	-	Casters, Rigid and Swivel (Institutional Duty)
HH-I-524	-	Insulation Board, Thermal (Polystyrene)
QQ-S-698	-	Steel, Sheet and Strip, Low Carbon
QQ-S-741	-	Steel, Carbon: Structural Shapes, Plates and Bars
WW-T-799	-	Tube, Copper, Seamless (for use with Solder-Type or Flared-Tube Fittings)
LLL-H-35	-	Hardboard, Fibrous-Felted (Fiberboard)

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MIL-R-3065 - Rubber, Fabricated Parts
MIL-R-12323 - Refrigerators and Related Equipment, Packaging and Packing of
MIL-R-20092 - Rubber (Synthetic), Sponge (Foamed Latex)

STANDARDS

FEDERAL

FED-STD-406 - Plastics: Methods of Testing

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130 - Identification Marking of U.S. Military Property
MIL-STD-417 - Rubber Compositions, Vulcanized General Purpose, Solid (Symbols and Tests)

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Underwriters' Laboratories, Incorporated (UL)

Standard No. 471 - Commercial Refrigerators

Electrical Appliance and Utilization Equipment List

(Application for copies should be addressed to Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; 333 Pfigsten Road, Northbrook, IL 60062; Walt Whitman Road, Melville, L.I., NY 11746; or 1655 Scott Blvd., Santa Clara, CA 95050.)

American Society of Heating, Refrigerating, and Air Conditioning Engineers, Inc. (ASHRAE)

Standard No. 13 - Methods of Rating and Testing Home Freezers

(Application for copies should be addressed to American Society of Heating, Refrigerating, and Air Conditioning Engineers, Inc., 345 East 47th Street, New York, NY 10017.)

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American Society for Testing and Materials (ASTM)

A 167	-	Corrosion-Resisting Chromium Nickel Steel, Plate, Sheet, and Strip
D 395	-	Compression Set of Vulcanized Rubber
B 86	-	Zinc Base Alloy Die Castings
D 883	-	Definition of Terms Relating to Plastics
D 1248	-	Polyethylene Molding and Extrusion Materials (Tentative)
D 2287	-	Nonrigid Vinyl Chloride Polymer and Copolymer Molding and Extrusion Compounds

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Design. The refrigerators shall be of two types, type I - Chest type and type II - Upright type. The refrigerators shall be designed to automatically maintain a selected storage compartment temperature between minus 5 degrees Fahrenheit (F) and plus 5 F. (see 3.1.1) by means of a mechanical operated, self-contained refrigeration system. The dimensions and capacities of each type refrigerator shall be as specified in table I. Whenever a material, design or construction change or deviation is made to the refrigeration unit during fulfillment of a contract, the Government shall stop acceptance of subsequent units until a sample unit containing the change or deviation has been tested or the supplier has submitted satisfactory evidence to the contracting officer indicating that the end product quality and performance characteristics have not been affected as a result of change or deviation.

TABLE I. Dimensions and capacities

Type	Storage capacities (cubic feet)	Length (inches)	Width (inches)	Depth (inches)	Height (inches)
I	11-13	(60-1/2) -67	(28-1/2) -32	-	39-42
II	15-20		52-1/2 (max)	28 (max)	73 (max)

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3.1.1 Performance. The refrigerators shall pass the performance tests specified in 4.5.2.4, except the unit shall attain a temperature of minus 20 F. in not more than eight hours of operation for no load pull down. The compressor running time for the no load test shall not exceed 75 percent of the total test time and the compressor running time for the storage load test shall not exceed 80 percent of the test time.

3.1.2 Operation. The refrigerators shall be tested as specified in 4.5.2.5 to determine that the refrigerators operate properly and that the refrigeration system cycles properly. There shall be no refrigerant leaks and the moisture content of the refrigerant at 100 F. shall not exceed 10 parts per million for the type 12 refrigerant and 45 parts per million for the type 22 refrigerant after completion of test specified in 4.5.2.5.

3.2 Standard product. The refrigerators covered by this specification shall be the current standard product of the manufacturer except for any design modifications specified herein.

3.3 Safety standard. The refrigerators shall conform to the applicable requirements of UL Standard No. 471.

* 3.3.1 Certification. The supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the refrigerators he proposes to furnish under this specification meet the applicable requirements of UL Standard No. 471. Acceptable evidence of meeting this requirement shall be the UL label, a listing in the "Electrical Appliance and Utilization Equipment List", or a certified test report acceptable to the Government from a recognized independent testing laboratory indicating that the refrigerators have been tested and conform to the applicable requirements of UL Standard No. 471.

3.4 Materials and components. Materials not definitely specified shall be of the quality normally used by the manufacturer for biological refrigerators provided the completed item complies with all provisions of this specification.

3.4.1 Carbon steel.

3.4.1.1 Sheet and strip. Carbon steel sheet and strip shall conform to QQ-S-698.

3.4.1.2 Structural. Carbon steel structural plates, shapes and bar shall conform to QQ-S-741.

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3.4.2 Stainless steel. Stainless steel plate, sheet, strip, bars, shapes, and forgings shall conform to the chemical composition of types 302, 304, 316, of ASTM A 167.

3.4.3 Zinc, die cast. High pressure die cast zinc shall conform to AG40A or AC41A of ASTM B 86.

3.4.4 Rubber.

3.4.4.1 Extruded rubber. Extruded rubber shall conform to SC 510 A_1F_1, of MIL-R-3065 and MIL-STD-417.

3.4.4.2 Sponge rubber. The sponge rubber shall conform to SC 510 A_1F_1, of MIL-R-3065 and MIL-STD-417 for the outer cover and type II, class 1 or 2, condition medium of MIL-R-20092 for the core.

3.4.5 Composition board. Composition board shall conform to type II of LLL-H-35.

3.4.6 Plastics.

3.4.6.1 Polyethylene. Thermoplastic plastics shall be virgin material except for spurs and runners from the same molding operation. Polyethylene plastics shall conform to type I of ASTM D 1248.

* 3.4.6.2 Styrene. Thermoplastic plastics shall be virgin material except for reclaimed spurs and runners from the same molding operation. Styrene plastics shall conform to the definition contained in ASTM D 883 for styrene rubber plastics. When styrene plastic is used for lid or door liner supports, it shall have a minimum Izod impact resistance of 0.7 foot pounds (ft lb) when tested as specified in 4.5.1.1. When styrene plastic is used for other components of the refrigerator it shall have a minimum Izod impact resistance of 0.25 ft lb when tested as specified in 4.5.1.1.

3.4.6.3 Vinyl chloride. Thermoplastic plastics shall be virgin material except for reclaimed spurs and runners from the same moulding operation. Vinyl chloride plastics shall be standard commercial types except that plasticizers shall be phthalate types only. Vinyl chloride plastics shall have a compression set of less than 20 percent when tested as specified in 4.5.1.4; a minimum tensile strength of 1100 pounds per square inch (psi), with minimum elongation at rupture of 250 percent when tested as specified in 4.5.1.2 and a weight loss not exceeding 2-1/2 percent when tested as specified in 4.5.1.3.

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3.4.6.4 Thermosetting. Thermosetting plastics shall be phenolic, melamine plastic, or reinforced polyesters. Molded parts shall have a flexural strength not less than 7500 psi and laminated parts not less than 10000 psi when tested as specified in 4.5.1.5. The maximum water absorption shall be 1.0 percent for molded parts and 4.0 percent for laminated parts when tested as specified in 4.5.1.6.

* 3.4.7 Refrigerants. The refrigerants shall conform to type 12 or type 22 of BB-F-1421.

3.4.8 Copper tube. Hard copper tube shall be seamless conforming to type K or L of WW-T-799. Soft copper tube shall be seamless, standard commercial refrigeration tube.

3.4.9 Insulation. Insulation shall be nonmoisture absorbing material; shall be homogenous throughout; and shall have manufacturers rated "K" factors specified below using a 75 F mean temperature and expressed in terms of British thermal units per hour (Btuh) per square foot per inch of thickness per degree F temperature difference. Loose fill type insulation will not be acceptable.

3.4.9.1 Pre-expanded insulation. Pre-expanded insulation shall be polystyrene material conforming to class 2, types I or II of HH-I-524.

* 3.4.9.2 Foamed-in-place insulation. Foamed-in-place insulation shall be polyurethane material having a manufacturer's commercial rated "K" factor of not more than 0.15, a core density of not less than 1.7 pounds per cubic foot when tested as specified in 4.5.1.7.

3.4.9.3 Semi-rigid. Semi-rigid insulation shall be "B" fiber or finer industrial type fiberglas with a manufacturer's commercial rated "K" factor of not more than 0.250 with a manufacturer's commercial rated density of at least 1.0 but not more than 3.0 pounds per cubic foot and shall be held together with a thermosetting binder.

3.5 Construction.

3.5.1 Cabinet. The type I cabinet shall be of the chest type with a single top lid. The interior shall be unobstructed by permanently mounted dividers or other parts. An opening shall be provided at the rear of the cabinet for inspection of an expansion valve and liquid and suction refrigerant lines. A plate, sealed vapor tight with a gasket and mastic cement, shall cover the opening. The type II cabinet shall be an upright

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type with two front doors of equal size. Each door shall provide access to one-half of the cabinet with each half of the cabinet divided into three compartments of equal size. Each compartment shall have a minimum of two adjustable shelves and an inner door shall be hinged at the bottom of each compartment so that the door can be opened and secured in a horizontal position. All tubes shall be supported. Tubes not inclosed within the outer shell of the cabinet shall be concealed in a channel or housing. Material for the channel or housing shall be equal in appearance to the outer shell of the cabinet. The refrigerator shall show no damage or permanent deformation of any part of the cabinet base or refrigeration system when tilted at an angle of 45 degrees as specified in 4.5.2.1 and when loaded and dropped as specified in 4.5.2.2. Tapping plates shall be furnished where necessary. The temperature of the outer shell shall not be more than 10 F below the ambient temperature when the refrigeration system is in operation and the storage compartment is at control temperature (see 4.5.2.4).

3.5.1.1 Machinery compartment. A machinery compartment for the condensing unit shall be provided and located at one end of the cabinet for the type I and located within the cabinet beneath the storage space for the type II. The machinery compartment shall be accessible at the front or rear of the cabinet through removable access panels. The access panels shall be louvered to provide ventilation. The access panels shall be attached to the cabinet by screw type fasteners that require a screwdriver to tighten and loosen.

* 3.5.1.2 Outer shells. The outer shell shall be fabricated from stainless steel specified in 3.4.2, and shall have a nominal thickness of 0.050 inch. The bottom of the outer shell may be a metal other than stainless steel provided the metal is corrosion-resistant or has been treated on both sides. When a metal other than stainless steel is used for the bottom, it shall not be visible from the front, rear, or sides of the refrigerator. Corners and edges on the outer shell shall be rounded. All seams shall be formed to eliminate exposed fasteners. Interior seam edges shall be coated with a sealing compound to assure a vaportight seal.

3.5.1.3 Inner liner. The inner liner shall be fabricated from stainless steel specified in 3.4.2 and shall have a nominal thickness of 0.0375 inch. The inner liner bottom shall be attached to the inner liner sides and back by spot or seam-welding. The interior of the cabinet shall not be cracked or have voids or imperfections. Inner liner corners shall be rounded and connecting joints shall be filled with solder.

* 3.5.1.3.1 Inner liner supports. The inner liner shall be joined to the outer shell by liner supports around the lid opening. The liner supports shall be fabricated from material specified in 3.4.5, 3.4.6.2, or 3.4.6.3.

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Inner liner supports shall be attached with stainless steel screws or chromium or nickel-plated brass screws. A secondary molded or extruded plastic inner liner support or breaker strip to cover the rigid liner support may be used as molding. The supports shall be sealed to the inner liner and outer shell with a waterproof mastic.

* 3.5.1.4 Lid for type I. The sides and top of the lid shall be fabricated from stainless steel specified in 3.4.2 and shall be one-piece with a nominal thickness of 0.050 inch. The inner liner of the lid shall be fabricated from material specified in 3.4.6.2 or 3.4.6.3 and shall be a nominal 0.060 inch thick. The lid shall have a gasket, fabricated from material specified in 3.4.4.1, and 3.4.4.2 or 3.4.6.3, attached to the inside periphery. The gasket shall be mounted to permit replacement. Air vents shall be provided in the lid liner to insure positive migration of any condensation in the insulation unless foamed-in-place insulation is used.

3.5.1.5 Doors for type II.

* 3.5.1.5.1 Exterior doors for type II. The outer shell of the exterior doors shall be fabricated from stainless steel specified in 3.4.2 and shall have a nominal thickness of 0.050 inch. All door corners shall be welded, ground, and polished to eliminate all traces of seams. The inner liner of the doors shall be fabricated from the material specified in 3.4.6.2 or 3.4.6.3 and shall have a nominal thickness of 0.060 inch. A gasket fabricated from material specified in 3.4.4.1 and 3.4.4.2 or 3.4.6.3 shall be attached to the inner liner periphery of the door. The gasket shall be mounted to permit replacement. Air vents shall be provided in the door inner liner to insure positive migration of any condensation contained in the door insulation.

* 3.5.1.5.2 Interior doors. The interior compartment doors for the type II refrigerators shall be fabricated from stainless steel specified in 3.4.2. Hinges and closing devices shall be fabricated from stainless steel specified in 3.4.2.

* 3.5.1.6 Hinges. The hinges shall be fabricated from copper-nickel chromium plated or nickel-chromium plated forged brass. The hinges for the type I refrigerators shall have fixed pins and shall be secured by fastening devices which cannot be removed while the lid is closed. The hinges shall be mounted on the rear of the cabinet and shall be spring loaded or counterbalanced to hold the lid in a raised position when open. When a spring loaded hinge is used the spring tension shall be adjustable. The hinges for the type II refrigerators shall be adjustable to compensate for a minimum 20 percent reduction of original gasket thickness on the hinged side of the door.

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* 3.5.1.7 Latch and striker. The latch and striker shall be fabricated from stainless steel; copper-nickel, chromium plated, malleable iron; nickel chromium plated cast or forged brass; or copper-nickel, chromium plated material specified in 3.4.3. The latch and strike assembly shall be adjustable to compensate for a minimum 20 percent reduction of the original gasket thickness on the latch side. Each handle shall have a built in locking device. The latch and striker shall be constructed to prevent a locked lid (type I) or door (type II) from being opened without removing or opening the lock.

3.5.1.8 Lid or door closure. The lid or door hinges and latches shall effect a seal that shall not allow the free insertion of a 0.003-inch gage, approximately 0.5 inch wide, between the gasket and the seating face when tested as specified in 4.5.2.3.

3.5.2 Base. The base shall be fabricated from steel specified in 3.4.1.1 or 3.4.1.2. The entire base shall be welded into a rigid structure and fastened to the cabinet by welding or bolting. The base shall be treated to resist corrosion.

3.5.2.1 Leveling devices. A leveling device shall be attached to each of the front corners of the base of the type II refrigerators to compensate for uneven floors.

* 3.5.2.2 Casters. For the type I refrigerators four or six casters, conforming to type II, class B, style c, mounting P, 5-inch diameter of FF-C-77, shall be attached to the bottom of the base. When four casters are used they shall be mounted one at each corner of the cabinet. When six casters are used they shall be mounted one at each corner of the cabinet and one at each corner of the machinery compartment. The casters shall be capable of passing the test specified in 4.5.2.6.

3.5.3 Installation of insulation. The refrigerators shall be insulated with material specified in 3.4.9.1, 3.4.9.2, or 3.4.9.3. The insulation thickness shall be determined by the manufacturer so that the computed overall heat transfer coefficient, "U", is not more than 0.10 Btuh per square foot per degree F for the top, four sides, lid (type I), and doors (type II), and not more than 0.07 Btuh per square foot, per degree F for the bottom. The computed "U" value shall be the "K" of the insulation value divided by the thickness of the insulation used. Rigid insulation, preformed, or foamed-in-place insulation shall be used for the bottom of the refrigerators. When a semirigid insulation is used for the top and four sides it shall be cut slightly larger and thicker than the shape and space to be filled so as to require forcing into place.

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3.5.4 Warning device. Unless otherwise specified (see 6.2), the refrigerator shall be equipped with a battery operated light and buzzer set to give warning signals when the cabinet temperature rises above the temperature set on temperature control, and shall be tested as specified in 4.5.2.5. The warning devices shall be capable of being manually de-energized when the cabinet is not in use. A means shall be provided for checking the operation of the warning device without interrupting the operation of the refrigerator.

3.5.5 Corrosion protection. Where dissimilar metals are to be joined a barrier material shall be used between such metals to prevent electrolytic or galvanic corrosion. Metal fasteners or fittings shall be of the same material as the material being joined or supported.

3.5.6 Electrical requirements. Unless otherwise specified (see 6.2), the refrigerator shall be designed for operation on a nominal 120-volt, 60-hertz, (hz) 1-phase, system. The refrigerators shall be completely electrically wired. When permitted by UL Standard No. 471, a power supply cord shall be furnished as part of the refrigerator. The cord shall have an attachment plug and shall extend at least 5-feet, but not more than 9-feet, beyond the point at which the cord is secured to the back of the cabinet. Where power supply cords are prohibited by UL Standard No. 471, the refrigerator shall be provided with inclosed terminal blocks for connection to external circuits.

3.5.7 Refrigeration system.

3.5.7.1 Evaporator. The evaporator for the type I shall be copper tube fastened to the sides, ends, and bottom of the insulation side of the inner liner or refrigerated plates. A thermal mastic or asphalt material shall be used to attach the tube on the liner to insure maximum efficiency. The tube shall not be placed where it will cause frost accumulation on the breaker strips. The evaporator for the type II shall be tube fastened to the underside of the top of each compartment; tube fastened to the sides, ends, and bottom, or top of each compartment on the insulation side of the inner liner; or plate coils forming the sides, ends, bottom, or top of each compartment. When tube is used for the evaporator on the insulation side, it shall be located and fastened in the same manner as the type I evaporators.

3.5.7.2 Condensing units.

3.5.7.2.1 Compressor. The compressor shall be hermetic type, thermally or mechanically sealed, except that for 50 hz operation open type condensing units shall be furnished.

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3.5.7.2.2 Condenser. The condenser shall be finned tube type or a plate coil type both employing natural convection. When spacing of the refrigerator from the wall is required to insure proper ventilation, spacer bolts or a duct shall be provided. Condensers shall be readily accessible for cleaning.

3.5.7.2.3 Compressor motor. The electric motor driving the compressor shall have sufficient horsepower to meet the performance and operation requirements of this specification. The motor shall be protected against overload by a thermal device with an automatic reset.

3.5.7.2.4 Heat exchanger. A heat exchanger shall be provided and installed.

3.5.7.2.5 Dehydrator. A dehydrator shall be provided and installed in the liquid refrigerant line.

3.5.7.2.6 Combination liquid and moisture indicator. A combination liquid and moisture indicator shall be provided and installed in the liquid refrigerant line.

3.5.7.2.7 Defrost system for type II. A semi-automatic, manually initiated electric or hot gas defrost system shall be provided.

3.5.7.3 Temperature control. The temperature control shall be a thermostatic type control. The temperature sensing element shall be located to be readily accessible and shall be easily removed. The control shall be adjustable to maintain temperatures selected between minus 5 F and plus 5 F.

3.5.7.4 Thermometer. The refrigerator shall be equipped with a dial type thermometer graduated in degrees Fahrenheit from minus 40 F to plus 40 F and marked with a safety zone from minus 20 F to plus 20 F to sense the interior temperature of the refrigerator. The thermometer shall be flush mounted on the outside front of the refrigerator in a visible location.

3.5.7.5 Refrigerant charge. The assembled refrigeration system shall be completely dehydrated and charged with either proper refrigerant specified in 3.4.7 and oil necessary for operation.

3.5.8 Lighting. The type II refrigerator shall be provided with interior illumination for all compartments. The lights shall be located within the cabinet and shall be recessed or protected by a cover. A switch for lights shall be operated automatically by opening the door, except multiple compartment doors may use a single pole tumbler switch, with integral red pilot light, with both the switch and the pilot light located in the front outside of the cabinet.

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* 3.5.9 Shelves. The shelves for the compartments of the type II refrigerators shall be fabricated from stainless steel wire. Shelves shall be the full width and depth of the compartment and shall be adjustable and removable. The increment for shelf adjustment shall be not more than 1 inch.

* 3.6 Finish. Stainless steel surfaces shall be given a No. 4 standard finish.

3.7 Marking. The type I and type II refrigerators shall be marked in accordance with MIL-STD-130.

3.7.1 Wiring diagram. A wiring diagram shall be affixed to the back of the cabinet or the condenser duct. The diagram shall show each electrical component and its terminal by sketch or outline and name, and not by electrical symbol alone. The diagram shall show the wiring of the refrigerator with the wires coded for identification.

3.8 Workmanship. The finished refrigerator shall be clean and all forming and cutting done accurately and clearly. All scale or flux deposits shall be removed from the finished welds. All exposed welds shall be ground smooth to match the adjacent surfaces. All nuts, bolts, and screws shall be tight.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

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4.3. In-process inspection.

4.3.1 Process examination. Examination shall be made of the following processes to establish that no deviations are made from the specified requirements. Whenever nonconformance is noted, correction shall be made to the process and items incorrectly processed.

(a) Sealing of interior outer shell seams (see 3.5.1.2)
(b) Corrosion-resistant treatment of base (see 3.5.2)
(c) Installation of insulation (see 3.5.3)
(d) Installation of tube (see 3.5.1)
(e) Corrosion protection (see 3.5.5)
(f) Attaching of tube (see 3.5.7.1)
(g) Charging of refrigeration system (see 3.5.7.5)

* 4.3.2 Intermediate testing.

* 4.3.2.1 Plastics. The plastics test shall be performed as specified in 4.5.1.1 through 4.5.1.6. The inspection lot shall be the amount of each type of plastic offered for inspection at one time. Any nonconformance shall be cause for rejection of the lot.

* 4.3.2.2 Insulation density test. The density tests of foamed-in-place insulation shall be performed as specified in 4.5.1.7. The inspection lot shall be the amount of material used in one day's production. The sample unit shall be a 6-inch cube.

4.3.3 End item inspection. The inspection lot shall consist of all refrigerators of one type offered for inspection at one time. The sampling unit for the inspection shall be one refrigerator.

4.3.3.1 Visual examination. Examination shall be made of the refrigerators for defects set forth in table II. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major and 6.5 for total defects expressed in terms of defects per hundred units.

TABLE II. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not No. 4 finish		X
	Stained		X

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TABLE II. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Construction and workmanship (applicable to type I and II)	Machinery compartment not located as specified	X	
	Corners and edges not rounded on outer shell	X	
	Exposed fasteners on outer shell		X
	Voids or cracks in inner liner	X	
	No gasket on lid or door, as applicable	X	
	Lid or door gasket not removable, as applicable		X
	No air vents in lid or exterior door inner liners, as applicable	X	
	Latch and striker assembly not adjustable		X
	No provision for locking on latch or strike assembly		X
	No power supply cord or inclosed terminal blocks (as applicable)	X	
	Temperature control not adjustable or not marked as specified	X	
	Temperature control not type specified		X
	Thermometer not dial type, not mounted or marked as specified		X
Type I	Cabinet interior obstructed by dividers or partitions		X
	No plate covered opening at rear of cabinet	X	
	Machinery compartment not accessible as specified or panels not removable or louvered		X
	Machinery compartment panels not attached as specified		X
	Lid hinge removable when lid is closed	X	
	Hinges not mounted as specified or not counter-balanced or spring loaded		X
	Hinge spring tension not adjustable (when applicable)		X
	Lid can be opened when bolts or screws securing strike are removed		X
	No casters	X	
	Incorrect number of casters		X
	Casters not type specified or not located as specified		X

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TABLE II. Classification of defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Construction and workmanship (applicable to type I and II) (continued)			
Type II	Not compartmented	X	
	Not compartmented as specified		X
	No shelves	X	
	Shelves not sized or adjustable as specified		X
	No doors on compartments or doors not hinged as specified		X
	Machinery compartment panel not removable	X	
	Exterior door corners show visual evidence of seam		X
	Exterior door hinge or door latch and strike not adjustable		X
	No key locking device on exterior door handle		X
	No leveling devices	X	
	Incorrect number of leveling devices or not located as specified		X
	No defrost system	X	
Marking	Missing, incomplete, not as specified		X
	No wiring diagram or not as specified	X	

4.3.3.2 Dimensional examination. The refrigerators shall be examined for defects in dimensions. Any dimension not within specified requirements shall be classified a defect. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

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4.3.3.3 End item testing. Each type I and type II refrigerator shall be tested as specified in 4.5.2.1, 4.5.2.2, 4.5.2.3, and 4.5.2.5. Any nonconformance shall be cause for rejection of the unit. One representative production unit (see 6.3) shall be tested as specified in 4.5.2.4 and 4.5.2.6 and one representative production unit (see 6.3) of type I refrigerators shall be tested as specified in 4.5.2.6. Any nonconformance shall be cause for rejection of the lot.

4.3.4 Examination of preparation for delivery. An examination shall be made to determine that preservation and packaging, packing, and marking as required by section 5 of this specification are in accordance with MIL-R-12323.

* 4.4 Certification examination. Certificates of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the specification requirements shall be examined and validated.

4.5 Test methods.

4.5.1 Material tests.

* 4.5.1.1 Izod impact. Styrene plastic shall be tested for Izod impact resistance in accordance with method 1071 of FED-STD-406 to determine compliance with 3.4.6.2. Failure to meet the requirements of 3.4.6.2 shall constitute failure of this test.

* 4.5.1.2 Tensile strength. Vinyl chloride plastic shall be tested for tensile strength in accordance with method 1011 of FED-STD-406 to determine compliance with 3.4.6.3. Failure to meet the requirements of 3.4.6.3 shall constitute failure of this test.

* 4.5.1.3 Weight loss. Vinyl chloride shall be tested for weight loss in accordance with test 7K of ASTM D 2287 to determine compliance with 3.4.6.3. Failure to meet the requirements of 3.4.6.3 shall constitute failure of this test.

* 4.5.1.4 Compression set test. A 1-inch section of the vinyl chloride shall be compressed in the direction of normal use to 60 percent of the original free height in a device described in method B ASTM D 395 and held for 7 days at 70 F. The free height shall be measured 30 minutes after removal to determine compliance with 3.4.6.3. Failure to meet the requirements of 3.4.6.3 shall constitute failure of this test.

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* 4.5.1.5 Flexural test. Thermosetting plastics shall be tested for flexural strength in accordance with method 1031 of FED-STD-406 to determine compliance with 3.4.6.3. Failure to meet the requirements of 3.4.6.3 shall constitute failure of this test.

* 4.5.1.6 Water absorption test. Thermosetting plastic shall be tested for water absorption in accordance with method 7031 of FED-STD-406 to determine compliance with 3.4.6.3. Failure to meet the requirements of 3.4.6.3 shall constitute failure of this test.

* 4.5.1.7 Density test for insulation. Prior to each day's production, one free foam sample of insulation specified in 3.4.9.2, a minimum size 8-inch cube shall be taken. From the center of the sample a 6-inch cube shall be cut and weighed. This weight, 1/8 of a cubic foot, shall be multiplied by 8 to determine conformance with the density requirements of 3.4.9.2. In addition a daily record of the foam insulation shall be kept as follows and made available to the contracting officer:

Calibration of mix (mix proportions)
Amount of flow (per box)
Time of flow (per box)
Temperature of mould

Failure to meet the requirements of 3.4.9.2 shall constitute failure of this test.

4.5.2 End products test.

* 4.5.2.1 Sturdiness test. The complete refrigerator shall be raised on one side bottom edge so that the bottom of the base makes an angle of 45 degrees with the horizontal to determine compliance with 3.5.1. The test shall be repeated so that the opposite side shall bear the load. At the completion of the test, the test specified in 4.5.2.3 will be performed. Failure to meet the requirements of 4.5.2.3 shall constitute failure of this test.

* 4.5.2.2 Drop test. The complete refrigerator shall be loaded to 50 percent of normal capacity by volume with suitable materials, not in liquid form, having an overall density of 35 pounds per cubic foot to determine compliance with 3.5.1. The material used for loading shall be nonrigid and shall be distributed uniformly over the bottom liner of the type I refrigerator and in the compartment of the type II refrigerator. Leveling devices

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shall be extended one-fourth of an inch beyond the plane of the lowest part of the type II refrigerator. One side of the refrigerator at a time shall then be raised three inches off a level concrete floor and be allowed to fall free in the following order: left side, rear, right side, front. The refrigerator shall be held in the elevated position for the drop test by a single support which can be withdrawn so the refrigerator is not hampered in its free fall or cause to fall in a plane other than parallel to the specified direction of fall. Failure to meet the requirements of 3.5.6 shall constitute failure of this test.

* 4.5.2.3 Lid or door seal test. The lid or door of the refrigerator shall be closed. An attempt shall be made to force a 0.003 inch metal gage, approximately 0.5 inch wide, between the gasket and the seating face at several points around the periphery of the door to determine compliance with 3.5.1.8. If the configuration of the door or trim does not permit the gage to reach the gasket, a bright light shall be placed inside the refrigerator. An observation for light showing between the gasket and seating face shall then be made in a darkened room. Failure to meet the requirements of 3.5.1.8 shall constitute failure of this test.

* 4.5.2.4 Performance test. A no load pull down test, a no load test, and a storage load test shall be performed in accordance with ASHRAE Standard No. 13 to determine compliance with 3.1.1. The temperature of the outer shell shall be measured at five locations (front, back, top and sides) during the no load pull down test when the equilibrium temperature is reached to determine compliance with 3.1.1. Failure to meet the requirements of 3.1.1 or 3.6.1 shall constitute failure of this test.

* 4.5.2.5 Operation test. Refrigerators shall be operated long enough at a control setting within the operating range to determine that the refrigerator operates and cycles properly as specified in 3.1.2 and that the warning device operates as specified in 3.5.4. After completion of this portion of the test, and with the cabinet at 70 F, plus or minus 5 F, all refrigerant joints shall be tested for leaks in accordance with the manufacturer's standard method to determine compliance with 3.1.2. Where the combination moisture and liquid indicator is used, it shall indicate "dry" within the limits specified in 3.1.2 during and upon completion of this test. When there is no moisture indicator, the moisture content of the refrigerant shall be determined in accordance with the manufacturer's standard method to determine compliance with 3.1.2. Failure to meet the requirements of 3.1.2 shall constitute failure of this test.

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* 4.5.2.6 Caster and marking test. An evenly distributed load of 1000 pounds, not in liquid form, shall be placed in the storage compartment of a type I refrigerator. Casters to be tested shall be attached to the bottom of the refrigerator and the refrigerator rolled back and forth over a smooth, clean concrete floor for 2 minutes. During the test the casters shall be observed to swivel and roll smoothly. Following the test, the casters shall be examined for failure or damage and the floor examined for marking. Test shall be performed to determine compliance with 3.5.2.2. Failure to meet the requirements of 3.5.2.2 shall constitute failure of this test.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. Each refrigerator shall be preserved and packaged in accordance with the applicable requirements of MIL-R-12323.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

* 5.2.1 Levels A, B and C. Each refrigerator, preserved and packaged as specified in 5.1, shall be packed in accordance with the applicable requirements of MIL-R-12323. In addition, for levels A and B, the base of each shipping container shall be provided with three cross load-bearing members to provide a minimum of 1/2 inch clearance for casters.

* 5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with the applicable requirements of MIL-R-12323.

6. NOTES

6.1 Intended use. The biological refrigerators covered by this specification are intended to store blood serums, to preserve surgical and autopsy specimens, and to maintain in a frozen state, tissues from which virus isolation is attempted.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of refrigerator required (see 1.2).
(c) When warning device is not required (see 3.5.4).
(d) When electrical characteristics are other than specified (see 3.5.6).
(e) Selection of applicable levels of preservation and packaging and packing (see 5.1 and 5.2).

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6.3 End item tests. Performance tests specified in 4.5.2.4 through 4.5.2.6 should be limited to one successful demonstration of compliance to the requirements of this specification for each contract.

6.4 Contract data requirements. Any data items to be delivered under any contract for items covered by this specification should be specifically called for in the contract in accordance with the applicable regulations of the procuring activity.

6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

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Navy - YD
Air Force - 82

User activity:

Navy - MS

Preparing activity:

Army - GL

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE: MIL-R-3860E — REFRIGERATORS, MECHANICAL, BIOLOGICAL, LOW-TEMPERATURE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

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